Amazon EventBridge

Problem: Your application needs to react to things that Happen. an EC2 instance stops, a user signs up in Cognito, a Salesforce (3rd party) deal closes, a codepipeline build fails. These are events.

Without Eventbridge: Polling architecture

Application polls EC2 API every 30s → "any state changes?"
App polls Codepipeline every 60s → "any builds?"
Salesforce every 5 min → "any deals closed?"

Disadv: Delayed reactions
Multiple blank API calls as there is no change.
Tight coupling
No central event routing

With EventBridge: near real time reactions
↓ events deliver in <500 ms
Zero polling
Central routing

Event Bridge: is a fully managed, serverless event bus service that connects applications using events from AWS services, SaaS partners & your own applications — routing them to targets via rules.

Event: A JSON document describing something that happened.

Event Bus: The pipeline that receives & routes ~~bus~~ event

Rule: Matches events by pattern & sends them to to targets.

Target: Destination — SQS, Lambda, Stepfunctions, another account EventBridge bus

Schema: Definition of what an event looks like (field names / types)

Persistence: Like SNS, EventBridge routes events, it does not store them.

Ex of an event:

```
{ "source": "aws.ec2",          → who sent it
  "detail-type": "EC2 Instance State-change",   → what happened
  "detail": {                   → kind of description
      "instance-id": "i-xxxxxxxx",   ← the specifics
      "state": "stopped"        → what state it changed to
  },
  "region": "us-east-1"
}
```

⇒ EB v/s SNS, SQS: Need code

↓ receives events from AWS services automatically - no code needed to publish.

Event Bus Types:

① Default Event Bus: Automatically created in every AWS account — cannot be deleted.

↳ automatically receives events from AWS services.

↳ Use this default to react for: automation & monitoring

② Custom Event Bus: You create & name these one per application or domain

↳ receives events from your own application

↳ EventBridge API is used here.

↳ use this for building event-driven microservices architectures.

③ Partner Event Bus: Created automatically when you enable a SaaS partner integration

↳ receives events directly from 3rd-party SaaS applications.

↳ Ex: Salesforce, Datadog, Github, Zendesk ...

↳ can have over 100's of partners.

↳ use for reacting to SaaS events with polling or custom

integration. = Only EB handles SaaS event natively.
Not SNS, SQS.

Rule: Event pattern (condition to match) +
Target(s) (where to send matching events)

a JSON document, all incoming traffic events gets compared to this.

⇒ if matched → routed to target,
if not → ignored

ex:
```
{
  "source":
  "detail-type":
}
```

detailed ex:
```
{
  "source": "aws.ec2",
  "detail-type": ["EC2 Instance State-Change"]
  "detail": {
      "state": ["stopped", "terminated"]
  }
}
```
↓
OR logic